A Tale of Five Decoders.

David Ponarovsky

November 3, 2025

Introduction

Today:

- Recap, and were we stood before the summer.
- Simulations Results.
- Where we heading.

Map

Simulations

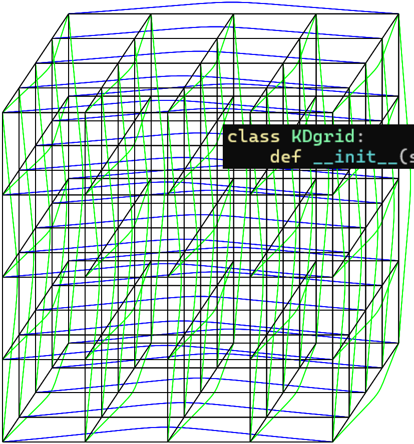
Perskil and Swift Rule?

Dennis at el?

Is the (d, k)-Toric a MEM?

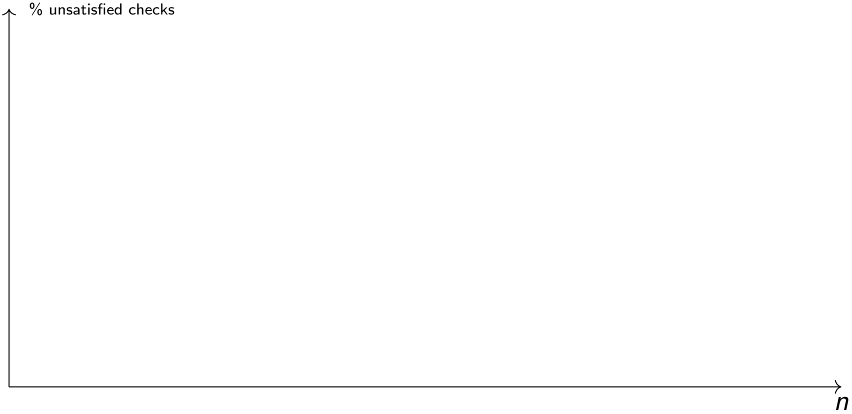
Coming with $O(1)$-depth gates for 'good' codes.

The Code



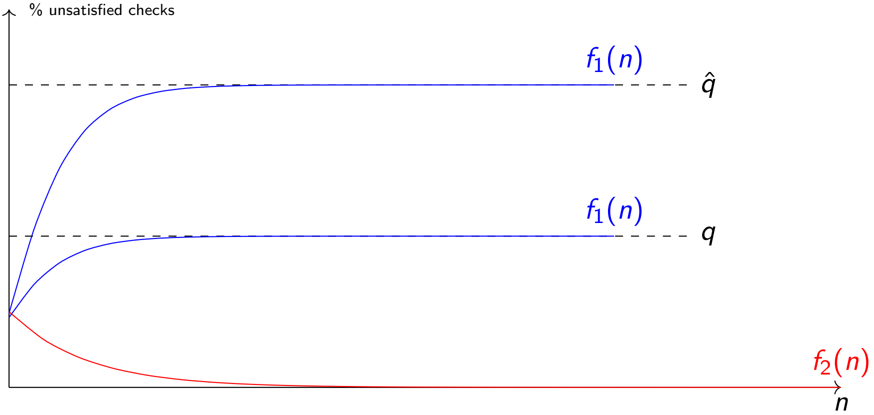
```python
class KDgrid:
    def __init__(self, n, k, celldim=2, checksdim =0,
```

How to Read



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Ideal Result.



colors

In red, No accumulation of errors. In blue decoding in the presence of noise.

Majority

Alg.

Each bit looks at its neighbour checks, if the majority of them is unsatisfied, then it flip itself.

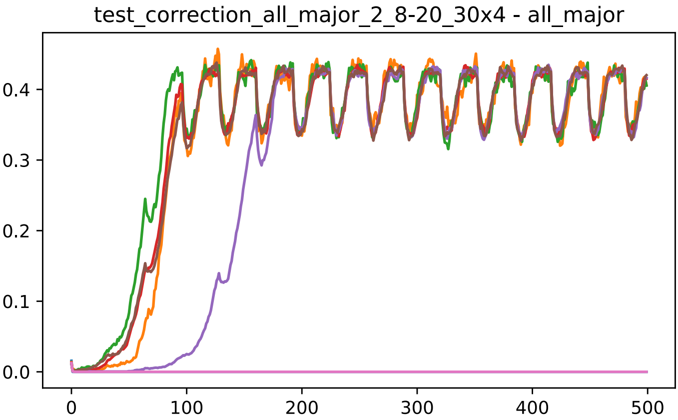
Justification.

Was suggested in Dennis.

Bottom line.

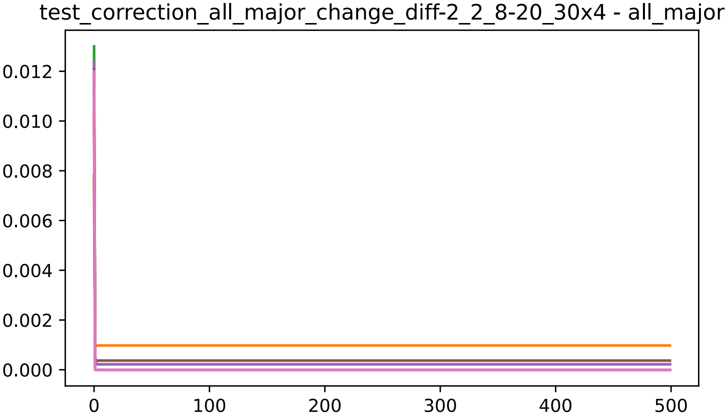
- Fails to correct, even in the absence of noise.
- Collisions.
- Gives better result when requiring $(\frac{1}{2} + \mu)$-majority.

Majority



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

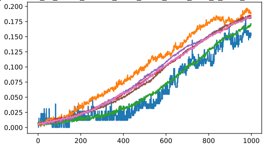
Majority



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

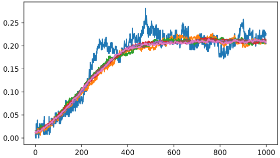
Majority

test_correction_all_major_change_diff-2_lower_noise_accu_2_8-20_30x4 - all_major



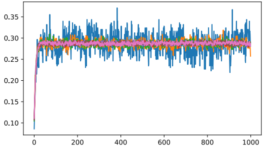
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.0005]
error accumulation :True

test_correction_all_major_change_diff-2_accu_2_8-20_30x4 - all_major



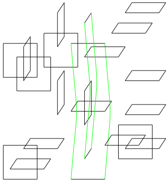
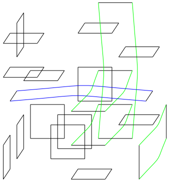
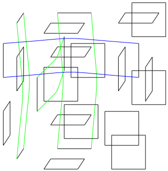
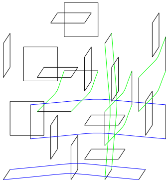
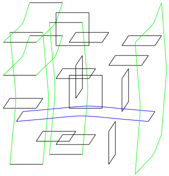
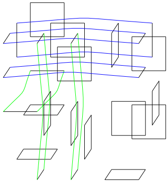
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :True

test_correction_all_major_change_diff-2_high_noise_accu_2_8-20_30x4 - all_major

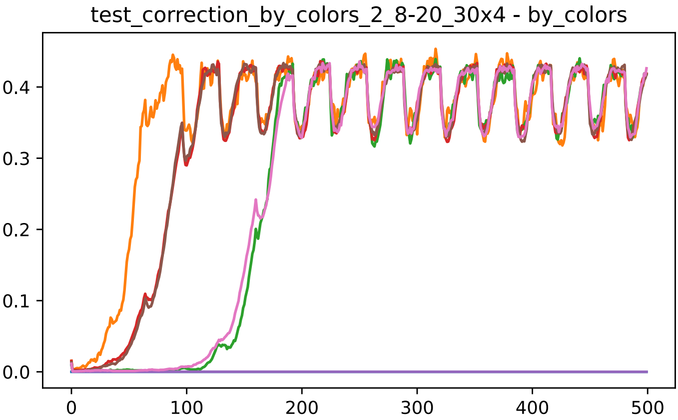


side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.01]
error accumulation :True

Colors

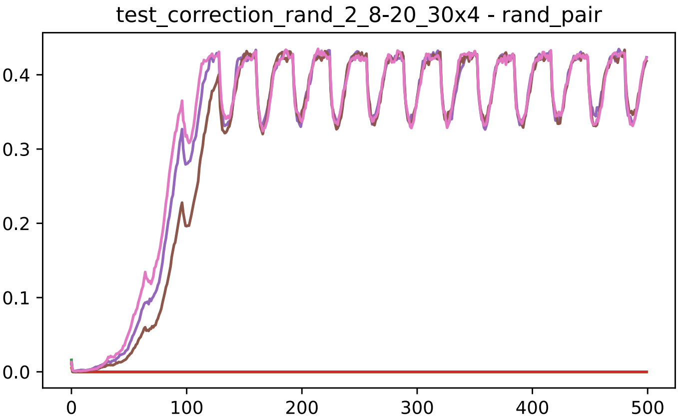







Colors



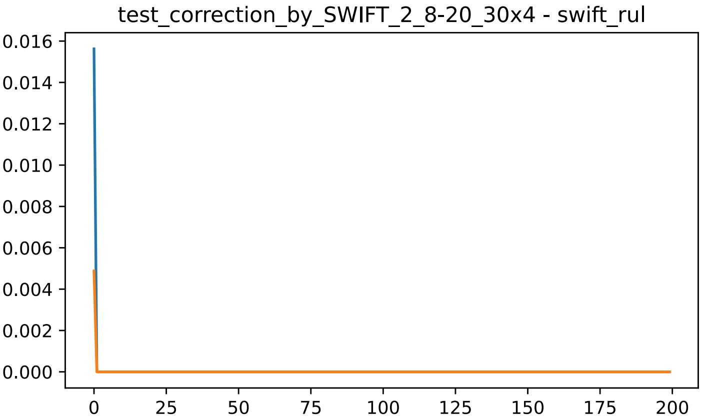
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Picking Random



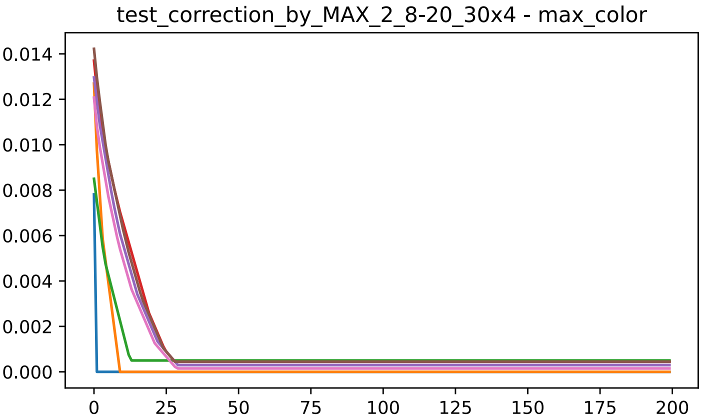
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

SWIFT



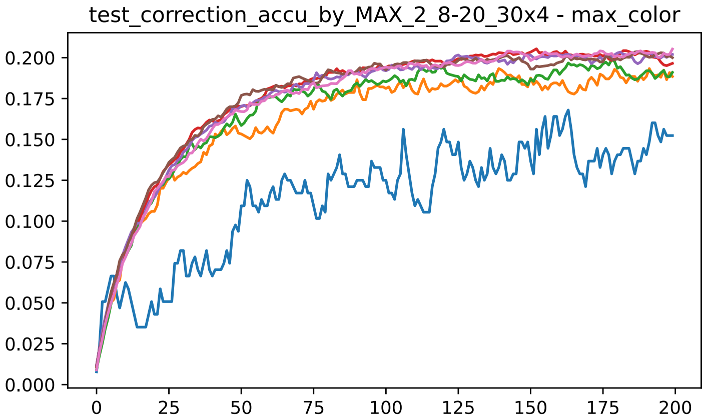
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Max Color



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Max Color



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :True